

Denoising Diffusion Probabilistic Models

A Detailed Mathematical and Conceptual Report

Prepared from the supplied DDPM study report

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Source and Scope

This report systematically reorganizes and explains the material contained in the supplied *Denoising Diffusion Probabilistic Model* document. The report follows its terminology, equations, conceptual distinctions, and derivations. In particular, it covers the forward and reverse Markov chains, the Gaussian noise schedule, the closed-form noising equation, latent-variable interpretation, variational inference and the ELBO, the tractable posterior, the noise-prediction parameterization, the simplified MSE objective, reverse sampling, the VAE connection, discretized decoding, score matching, Langevin dynamics, and the important training-versus-generation distinction.

The supplied document explicitly emphasizes that the forward process is fixed rather than learned, while the reverse process is the trainable generative component. It also emphasizes that the forward process is nevertheless essential because it creates the noisy training examples and the exact noise targets.

Chapter 1

Introduction to Diffusion Models

1.1 The central idea

Let x_0 denote a clean observation, such as an image. A DDPM constructs a sequence

$$x_0, x_1, x_2, \dots, x_T.$$

The forward direction gradually destroys information:

$$x_0 \longrightarrow x_1 \longrightarrow x_2 \longrightarrow \dots \longrightarrow x_T$$

where x_T is approximately standard Gaussian noise.

The generative direction reverses this chain:

$$x_T \longrightarrow x_{T-1} \longrightarrow \dots \longrightarrow x_1 \longrightarrow x_0.$$

The key conceptual statement is:

Define an easy process that turns data into Gaussian noise, then learn its reverse.

The forward process is known and fixed. The reverse process is learned.

1.2 Why the model is probabilistic

A transition is not written as a deterministic function

$$x_t = f(x_{t-1}),$$

but as a conditional distribution

$$x_t \sim q(x_t \mid x_{t-1}).$$

Thus every transition contains randomness. The forward chain is a Markov chain:

$$q(x_t \mid x_{0:t-1}) = q(x_t \mid x_{t-1}).$$

The Markov property means that once the current state x_{t-1} is known, earlier states are not needed to generate the next state.

1.3 Observed and latent variables

The observed data is x_0 . The intermediate states

$$x_1, \dots, x_T$$

are latent variables. Therefore DDPM is a latent-variable model with a long chain of latent variables rather than one compact latent vector.

The model marginalizes these latent variables:

$$p_\theta(x_0) = \int p_\theta(x_{0:T}) dx_{1:T}.$$

This is directly analogous to latent-variable models such as VAEs.

Chapter 2

The Forward Diffusion Process

2.1 Markov factorization

The forward process is

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^T q(x_t \mid x_{t-1}).$$

Each transition is Gaussian:

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I\right).$$

Define

$$\alpha_t = 1 - \beta_t.$$

Then

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{\alpha_t} x_{t-1}, (1 - \alpha_t)I\right).$$

2.2 Meaning of β_t

The scalar β_t is the variance of the new Gaussian noise introduced at timestep t . A schedule

$$\beta_1, \beta_2, \dots, \beta_T$$

is selected before training.

A small β_t means that the current step makes only a small change. Repeating many small changes gradually destroys the original signal.

2.3 Reparameterized form

Because

$$\epsilon \sim \mathcal{N}(0, I),$$

the Gaussian transition can be sampled as

$$x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon.$$

Equivalently,

$$x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1 - \alpha_t} \epsilon.$$

The square roots appear because variance is the square of standard deviation:

$$\text{std} = \sqrt{\text{Var}}.$$

2.4 Where does $\sqrt{1 - \beta_t}$ come from?

Start with the general form

$$x_t = a_t x_{t-1} + b_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

The variance contributed by the noise is b_t^2 . DDPM chooses

$$b_t^2 = \beta_t,$$

so

$$b_t = \sqrt{\beta_t}.$$

The signal and noise scaling are chosen to maintain a normalized overall scale:

$$a_t^2 + b_t^2 = 1.$$

Therefore

$$a_t^2 + \beta_t = 1$$

and hence

$$a_t = \sqrt{1 - \beta_t}.$$

This normalization is a design choice of the DDPM forward process; it is not a consequence that every Gaussian transition must satisfy.

2.5 Conditional randomness versus accumulated noise

When considering

$$q(x_t \mid x_{t-1}),$$

the value x_{t-1} is treated as fixed. The only new randomness in this conditional transition is the newly sampled ϵ_t .

However, x_{t-1} itself already contains noise from earlier steps:

$$x_{t-1} = \sqrt{1 - \beta_{t-1}} x_{t-2} + \sqrt{\beta_{t-1}} \epsilon_{t-1}.$$

Thus the correct interpretation is:

At each step, treat the current x_{t-1} as given and add new noise.

Chapter 3

The Closed-Form Noising Equation

3.1 Cumulative signal coefficient

Define

$$\bar{\alpha}_t = \prod_{s=1}^t \alpha_s.$$

Repeated substitution of the one-step transitions yields

$$q(x_t \mid x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)I).$$

Therefore we can sample x_t directly:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

3.2 Why this equation is fundamental

Without the closed form, producing x_t would require

$$x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_t.$$

The closed form permits a direct jump:

$$x_0 \longrightarrow x_t.$$

Thus training can choose a random timestep and immediately construct the corresponding noise level.

3.3 Signal-noise interpretation

The direct equation can be written as

$$x_t = \underbrace{\sqrt{\bar{\alpha}_t} x_0}_{\text{remaining signal}} + \underbrace{\sqrt{1 - \bar{\alpha}_t} \epsilon}_{\text{accumulated noise}}.$$

For small t ,

$$\bar{\alpha}_t \approx 1, \quad x_t \approx x_0.$$

For sufficiently large t ,

$$\bar{\alpha}_t \approx 0, \quad x_t \approx \epsilon,$$

so

$$x_T \approx \mathcal{N}(0, I).$$

For example, if

$$\bar{\alpha}_t = 0.25,$$

then

$$x_t = 0.5x_0 + \sqrt{0.75} \epsilon \approx 0.5x_0 + 0.866\epsilon.$$

The numerical coefficients should be interpreted as scaling coefficients in the signal-plus-noise construction rather than as literal percentages of image pixels.

Chapter 4

The Reverse Generative Process

4.1 Reverse joint distribution

The reverse model is defined as

$$p_{\theta}(x_{0:T}) = p(x_T) \prod_{t=1}^T p_{\theta}(x_{t-1} \mid x_t).$$

The terminal prior is

$$p(x_T) = \mathcal{N}(0, I).$$

Generation therefore starts with

$$x_T \sim \mathcal{N}(0, I)$$

and repeatedly samples

$$x_{t-1} \sim p_{\theta}(x_{t-1} \mid x_t).$$

4.2 Gaussian reverse transition

The reverse transition is modeled as

$$p_{\theta}(x_{t-1} \mid x_t) = \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t), \Sigma_{\theta}(x_t, t)).$$

In the original DDPM formulation emphasized in the supplied document, the covariance can be fixed:

$$\Sigma_{\theta}(x_t, t) = \sigma_t^2 I.$$

The network therefore primarily learns the information needed to determine the reverse mean.

4.3 Why the timestep is supplied to the network

The same noisy image representation can correspond to different denoising problems depending on t . At small t , the image may be nearly clean; at large t , it may be almost pure noise.

Therefore the network receives

$$(x_t, t).$$

A time embedding can encode the timestep so that the network knows the noise regime in which it is operating.

Chapter 5

Training: Why Variational Inference Appears

5.1 The likelihood objective

Ideally, maximum likelihood would maximize

$$\log p_{\theta}(x_0)$$

or equivalently minimize

$$-\log p_{\theta}(x_0).$$

But

$$p_{\theta}(x_0) = \int p_{\theta}(x_{0:T}) dx_{1:T}$$

contains a high-dimensional latent-variable integral, making direct likelihood optimization intractable.

5.2 Variational bound

The forward process

$$q(x_{1:T} \mid x_0)$$

acts as a fixed approximate posterior over the latent chain.

The variational objective has the form

$$\log p_{\theta}(x_0) \geq \mathbb{E}_q \left[\log \frac{p_{\theta}(x_{0:T})}{q(x_{1:T} \mid x_0)} \right].$$

Equivalently, in negative-log form,

$$-\log p_{\theta}(x_0) \leq L.$$

The important point is not that the bound is automatically the exact likelihood. Rather, it is a tractable surrogate that can be optimized. If the variational posterior matches the true posterior, the bound becomes tight.

5.3 Substitution of the two Markov chains

The reverse factorization is

$$p_{\theta}(x_{0:T}) = p(x_T) \prod_{t=1}^T p_{\theta}(x_{t-1} \mid x_t),$$

while

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^T q(x_t \mid x_{t-1}).$$

Substituting these expressions into the variational objective produces a sum of terms involving the terminal prior, forward transitions, and learned reverse transitions.

Chapter 6

ELBO Decomposition

The variational objective can be reorganized as

$$L = L_T + \sum_{t>1} L_{t-1} + L_0.$$

The terms are

$$L_T = D_{\text{KL}}(q(x_T \mid x_0) \parallel p(x_T))$$

$$L_{t-1} = D_{\text{KL}}(q(x_{t-1} \mid x_t, x_0) \parallel p_\theta(x_{t-1} \mid x_t))$$

and

$$L_0 = -\log p_\theta(x_0 \mid x_1).$$

6.1 L_T : matching the terminal prior

The forward process produces

$$q(x_T \mid x_0),$$

while generation begins from

$$p(x_T) = \mathcal{N}(0, I).$$

Thus

$$L_T = D_{\text{KL}}(q(x_T \mid x_0) \parallel p(x_T))$$

encourages the terminal forward distribution to agree with the Gaussian distribution used for generation.

For sufficiently large T ,

$$q(x_T \mid x_0) \approx \mathcal{N}(0, I),$$

so this term is small and, because the forward process is fixed, it is not a trainable neural-network term in the usual setup.

6.2 L_{t-1} : the central learning term

The intermediate objective is

$$L_{t-1} = D_{\text{KL}}(q(x_{t-1} \mid x_t, x_0) \parallel p_{\theta}(x_{t-1} \mid x_t)) .$$

The two distributions have different roles:

- $q(x_{t-1} \mid x_t, x_0)$ is the analytically tractable target posterior available during training.
- $p_{\theta}(x_{t-1} \mid x_t)$ is the learned reverse distribution that must work without x_0 .

Therefore training teaches

$$\boxed{p_{\theta}(x_{t-1} \mid x_t) \approx q(x_{t-1} \mid x_t, x_0)} .$$

Chapter 7

Why the True Posterior Contains x_0

7.1 The apparent paradox

The model ultimately needs a reverse distribution that operates from x_t alone:

$$q(x_{t-1} \mid x_t).$$

Yet the tractable training target is

$$q(x_{t-1} \mid x_t, x_0).$$

The reason is that conditioning on x_0 makes the posterior analytically tractable.

7.2 Why Gaussianity is not automatic in the reverse direction

The fact that

$$q(x_t \mid x_{t-1})$$

is Gaussian does not by itself imply

$$q(x_{t-1} \mid x_t)$$

is Gaussian.

Bayes' rule gives

$$q(x_{t-1} \mid x_t) \propto q(x_t \mid x_{t-1})q(x_{t-1}).$$

The marginal $q(x_{t-1})$ can be complicated.

However, conditioning on x_0 gives

$$q(x_{t-1} \mid x_t, x_0) \propto q(x_t \mid x_{t-1})q(x_{t-1} \mid x_0).$$

Both factors are Gaussian. Therefore

$q(x_{t-1} \mid x_t, x_0)$ is Gaussian.

This is the key reason x_0 appears in the variational decomposition.

Chapter 8

The Analytic Forward Posterior

The tractable posterior has the form

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\left(x_{t-1}; \tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I\right).$$

The posterior mean is

$$\tilde{\mu}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}x_0 + \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}x_t.$$

The posterior variance is

$$\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}\beta_t.$$

The important intuition is that the true posterior mean is a weighted combination of the clean image and the current noisy state:

$$\tilde{\mu}_t = w_1 x_0 + w_2 x_t.$$

During training, both x_0 and x_t are available, so this target distribution can be computed analytically.

Chapter 9

Gaussian KL and Rao–Blackwellization

9.1 Why Gaussian KL is convenient

For two Gaussian distributions,

$$q = \mathcal{N}(\mu_1, \Sigma_1), \quad p = \mathcal{N}(\mu_2, \Sigma_2),$$

the KL divergence has a closed form:

$$D_{\text{KL}}(q\|p) = \frac{1}{2} \left[\text{tr}(\Sigma_2^{-1}\Sigma_1) + (\mu_2 - \mu_1)^T \Sigma_2^{-1} (\mu_2 - \mu_1) - k + \log \frac{|\Sigma_2|}{|\Sigma_1|} \right].$$

Therefore the training objective does not need to estimate every KL term using high-variance Monte Carlo sampling.

The supplied report describes this as a Rao–Blackwellized treatment: when a conditional expectation or KL can be computed analytically, using that exact calculation reduces unnecessary sampling variance.

Chapter 10

What Is Actually Trained?

10.1 Forward process versus reverse process

The distinction is fundamental:

Forward process: fixed, known, not trained

Reverse process: learned by the neural network

The forward process is nevertheless essential for training.

A training example is generated as follows:

$$x_0 \longrightarrow x_t$$

using

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon.$$

Because ϵ was sampled by us, it is known exactly. The training pair is effectively

$$(x_t, t) \longrightarrow \epsilon.$$

Thus the fixed forward process creates the supervised-like denoising task, while the reverse network learns its solution.

10.2 Training algorithm

For each training example:

1. Sample a clean image x_0 .

2. Sample a timestep

$$t \sim \text{Uniform}\{1, \dots, T\}.$$

3. Sample Gaussian noise

$$\epsilon \sim \mathcal{N}(0, I).$$

4. Construct

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon.$$

5. Feed (x_t, t) to the neural network.

6. Obtain $\epsilon_\theta(x_t, t)$.

7. Compare predicted and true noise.

8. Update θ by gradient descent.

Chapter 11

Noise-Prediction Parameterization

11.1 From noisy image to clean image

Starting from

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon,$$

solve for x_0 :

$$x_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\epsilon}{\sqrt{\bar{\alpha}_t}}.$$

If the network predicts

$$\epsilon_\theta(x_t, t) \approx \epsilon,$$

then

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}.$$

Thus predicting noise implicitly gives information about the clean sample.

11.2 Three possible parameterizations

The report discusses three conceptual choices:

1. Predict the posterior mean:

$$\mu_\theta(x_t, t) \approx \tilde{\mu}_t.$$

2. Predict the clean data:

$$x_\theta(x_t, t) \approx x_0.$$

3. Predict the noise:

$$\epsilon_\theta(x_t, t) \approx \epsilon.$$

The original DDPM formulation found noise prediction particularly effective.

Chapter 12

Deriving the Learned Reverse Mean

The intermediate KL between two Gaussians reduces, up to constants, to a mean-matching objective:

$$L_{t-1} = \frac{1}{2\sigma_t^2} \mathbb{E} \left[\|\tilde{\mu}_t - \mu_\theta\|^2 \right] + C.$$

Substitute

$$x_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon}{\sqrt{\bar{\alpha}_t}}$$

into $\tilde{\mu}_t$. After simplification:

$$\tilde{\mu}_t = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right).$$

Now replace the unknown true noise with the network prediction:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right).$$

This yields the central computational chain:

$$(x_t, t) \rightarrow \epsilon_\theta \rightarrow \mu_\theta \rightarrow x_{t-1}.$$

The network therefore does not have to directly output x_{t-1} .

Chapter 13

The Simplified DDPM Loss

The exact variational objective produces a timestep-dependent weighted noise-prediction term:

$$L_{t-1} \propto w_t \|\epsilon - \epsilon_\theta(x_t, t)\|^2,$$

where

$$w_t = \frac{\beta_t^2}{2\sigma_t^2\alpha_t(1 - \bar{\alpha}_t)}.$$

The practical objective removes this weighting:

$$L_{\text{simple}} = \mathbb{E}_{x_0, t, \epsilon} \left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2 \right].$$

13.1 ELBO versus simplified loss

It is important not to identify the two expressions as mathematically identical:

$$L_{\text{simple}} \neq L_{\text{VLB}}$$

in the strict sense.

Rather, the simplified objective is derived from the variational formulation and uses a different weighting. The supplied document emphasizes that this simpler objective was found empirically useful for sample quality.

13.2 Why the objective resembles supervised learning

The target noise is known exactly:

$$\epsilon \sim \mathcal{N}(0, I).$$

The model receives

$$(x_t, t)$$

and tries to reproduce the sampled target:

$$\epsilon_\theta(x_t, t) \approx \epsilon.$$

Hence the optimization looks like ordinary regression, even though the resulting model is generative.

Chapter 14

Why Predicting Noise Teaches Image Structure

At first it may appear that the network only learns noise. However, the network sees the noisy image x_t , not the noise in isolation.

To identify the noise, it must learn which structures in x_t are likely to be signal and which are likely to be corruption.

Consequently it learns statistical regularities such as:

- edges,
- textures,
- shapes,
- object structure,
- spatial relationships,
- patterns that remain plausible across noise levels.

The network is therefore learning the structure of the data distribution through a denoising task.

Chapter 15

Reverse Sampling

After training, x_0 is unavailable.

Generation starts from

$$x_T \sim \mathcal{N}(0, I).$$

At each timestep:

$$x_{t-1} \sim p_\theta(x_{t-1} \mid x_t).$$

Using the Gaussian form:

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z, \quad z \sim \mathcal{N}(0, I).$$

Substituting the noise-prediction mean gives

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \alpha_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z.$$

The generation chain is therefore

$$\mathcal{N}(0, I) \rightarrow x_{T-1} \rightarrow x_{T-2} \rightarrow \cdots \rightarrow x_0.$$

15.1 Why is random noise added during sampling?

The reverse model represents a probability distribution, not merely a single deterministic point:

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(\mu_\theta, \sigma_t^2 I).$$

Taking only the mean would produce one deterministic representative. Sampling

$$x_{t-1} = \mu_\theta + \sigma_t z$$

samples from the modeled conditional distribution.

The randomness is therefore part of probabilistic generation rather than an error.

15.2 The last-step special case

At the final reverse step, the model must produce a discrete-valued image while Gaussian transitions are continuous.

For a pixel value such as 127, the likelihood is therefore based on probability mass in the corresponding bin, approximately

$$[126.5, 127.5],$$

rather than the Gaussian density at one exact point.

This is the purpose of the discretized Gaussian decoder in the final likelihood term.

Chapter 16

Training Versus Generation

Table 16.1: Training and generation in DDPM

Training	Generation
Starts with real x_0	Starts with random x_T
Uses the fixed forward process	Does not need the forward process
x_0 is available	x_0 is unknown
Constructs x_t directly	Iteratively produces x_{t-1}
Target noise ϵ is known	Noise is predicted by the network
Learns $\epsilon_\theta(x_t, t)$	Uses ϵ_θ to denoise
Optimizes the network parameters	Samples from the learned reverse chain

A compact comparison is:

Training: $x_0 \rightarrow x_t \rightarrow \epsilon_\theta$
Generation: $x_T \rightarrow x_{T-1} \rightarrow \cdots \rightarrow x_0$

Chapter 17

Why Many Small Noise Steps?

The diffusion process uses many small changes rather than one enormous corruption step.

A conceptual large transformation

image $\rightarrow$ pure noise

would be difficult to reverse directly.

Instead:

image $\rightarrow$ slightly noisy $\rightarrow$ more noisy $\rightarrow \dots \rightarrow$ noise.

The reverse problem then becomes a sequence of small denoising problems.

hard transformation $\approx$ many easier transformations.
--

The supplied report also explains why Gaussian reverse transitions are reasonable when the forward process uses small Gaussian perturbations: locally, the reverse conditional can be well approximated by a Gaussian.

Chapter 18

Connection to Variational Autoencoders

18.1 VAE structure

A VAE commonly has

$$x \rightarrow z \rightarrow x.$$

The encoder is

$$q_\phi(z \mid x)$$

and the decoder is

$$p_\theta(x \mid z).$$

The VAE uses the ELBO

$$\mathcal{L} = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{\text{KL}}(q_\phi(z \mid x) \| p(z)).$$

18.2 DDPM as a long latent-variable model

DDPM instead has

$$x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_T.$$

Thus:

$x_1, \dots, x_T$ are the latent variables.

The correspondence is:

Table 18.1: VAE and DDPM correspondence

VAE	DDPM
Observed data x	Observed data x_0
Latent z	Latent chain $x_1, \dots, x_T$
Encoder $q_\phi(z \mid x)$	Fixed forward process $q(x_{1:T} \mid x_0)$
Prior $p(z)$	$p(x_T) = \mathcal{N}(0, I)$
Decoder $p_\theta(x \mid z)$	Learned reverse chain $p_\theta(x_{0:T})$
ELBO	Diffusion variational bound

18.3 The crucial difference

In a standard VAE, the approximate posterior

$$q_{\phi}(z \mid x)$$

is learned.

In the original DDPM formulation,

$$q(x_{1:T} \mid x_0)$$

is fixed through the selected β_t schedule.

Thus the forward process is analogous to a VAE encoder in its variational role, but unlike a VAE encoder it has no learned parameters in the original DDPM construction.

Chapter 19

Markov Chains in Diffusion

A Markov chain satisfies

$$P(X_{t+1} \mid X_t, X_{t-1}, \dots, X_0) = P(X_{t+1} \mid X_t).$$

For diffusion:

$$\boxed{q(x_t \mid x_{t-1}, x_{t-2}, \dots, x_0) = q(x_t \mid x_{t-1}).}$$

This means the next forward state only needs the current state.

However, the Markov property must not be incorrectly reversed. It does not imply

$$q(x_{t-1} \mid x_t, x_0) = q(x_{t-1} \mid x_t).$$

The earlier observation x_0 can provide additional information about the hidden previous state when reasoning backward.

Chapter 20

KL Divergence and Its Non-Negativity

The KL divergence is

$$D_{\text{KL}}(q\|p) = \mathbb{E}_q \left[\log \frac{q(x)}{p(x)} \right].$$

The pointwise quantity

$$\log \frac{q(x)}{p(x)}$$

can be negative. Nevertheless,

$$\boxed{D_{\text{KL}}(q\|p) \geq 0.}$$

One derivation uses

$$\log r \leq r - 1, \quad r > 0.$$

Let

$$r(x) = \frac{p(x)}{q(x)}.$$

Then

$$-\log r \geq 1 - r.$$

Taking expectation under q :

$$-\mathbb{E}_q[\log r] \geq 1 - \mathbb{E}_q[r].$$

But

$$\mathbb{E}_q[r] = \int q(x) \frac{p(x)}{q(x)} dx = \int p(x) dx = 1.$$

Therefore

$$D_{\text{KL}}(q\|p) \geq 0.$$

The key distinction is:

$$\boxed{\log(q/p) \text{ need not be positive pointwise, but its expectation under } q \text{ is non-negative.}}$$

Chapter 21

Image Scaling and Discretized Likelihood

The supplied report describes scaling image pixels from

$$0, 1, \dots, 255$$

to

$$[-1, 1].$$

A standard transformation is

$$x_{\text{scaled}} = 2 \frac{x}{255} - 1.$$

Thus

$$0 \mapsto -1, \quad 255 \mapsto 1.$$

This is convenient because the terminal Gaussian prior is centered around zero, so the data representation is also centered around zero.

For the final decoder, however, observed pixels are discrete while Gaussian model outputs are continuous. The likelihood therefore integrates Gaussian probability over each quantization bin.

For a pixel value 127:

$$P(x_0^i = 127) \approx \int_{126.5}^{127.5} p(x) dx.$$

This produces a mathematically appropriate probability for discrete image observations.

Chapter 22

Score Matching Interpretation

22.1 The score

The score of a density $p(x)$ is

$$\nabla_x \log p(x).$$

It describes the direction in which the log-density increases.

For the noisy marginal $q_t(x_t)$, diffusion can be interpreted as learning

$$\nabla_{x_t} \log q_t(x_t).$$

Under the appropriate marginal interpretation, the noise predictor is related to the score:

$$\epsilon_\theta(x_t, t) \approx -\sqrt{1 - \bar{\alpha}_t} \nabla_{x_t} \log q_t(x_t).$$

Thus noise prediction and score estimation are closely connected.

22.2 Denoising score matching intuition

The denoising interpretation is:

Given this noisy point, which direction leads toward more likely data?

Different timesteps correspond to different noise scales. Hence one model learns denoising behavior across many scales:

$$t = 1, 2, \dots, T.$$

Chapter 23

Langevin Dynamics Connection

A simplified Langevin update has the form

$$x' = x + \eta \nabla_x \log p(x) + \sqrt{2\eta} z, \quad z \sim \mathcal{N}(0, I).$$

It contains two ingredients:

1. a directional movement toward high-density regions;
2. random perturbation.

The DDPM reverse update has the same conceptual structure:

$$x_{t-1} = \text{denoising direction} + \text{Gaussian randomness}.$$

The supplied report therefore presents two complementary interpretations:

Variational inference / latent-variable model

and

Denoising score matching / Langevin-like sampling.

These are different mathematical views of the same diffusion construction.

Chapter 24

Why Randomness Is Retained in Reverse Sampling

Suppose

$$p_{\theta}(x_{t-1} \mid x_t) = \mathcal{N}(\mu_{\theta}, \sigma_t^2 I).$$

A deterministic update would use only

$$x_{t-1} = \mu_{\theta}.$$

But the model represents a distribution, so sampling instead gives

$$x_{t-1} = \mu_{\theta} + \sigma_t z.$$

The random variable z allows the generated trajectory to explore different plausible reverse paths.

Thus two different initial Gaussian noise samples, and potentially different random variables at the reverse steps, can produce different samples from the learned data distribution.

Chapter 25

Complete End-to-End Mathematical Picture

The complete forward process is

$$\boxed{x_0 \xrightarrow[\text{fixed}]{\text{Gaussian corruption}} x_t}$$

with

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon.$$

During training:

$$\boxed{(x_t, t) \xrightarrow{\text{neural network}} \epsilon_\theta(x_t, t).}$$

The prediction gives an estimate of the clean sample:

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}.$$

The predicted noise determines the reverse mean:

$$\boxed{\mu_\theta = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right)}.$$

During generation:

$$\boxed{x_{t-1} = \mu_\theta + \sigma_t z, \quad z \sim \mathcal{N}(0, I).}$$

Finally:

$$\boxed{x_T \rightarrow x_{T-1} \rightarrow \cdots \rightarrow x_1 \rightarrow x_0.}$$

Chapter 26

The Seven Equations to Remember

The supplied material identifies a compact set of equations that captures most of DDPM:

1. Forward one-step transition

$$q(x_t \mid x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}x_{t-1}, \beta_t I).$$

2. Direct noising

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon.$$

3. True training posterior

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}(\tilde{\mu}_t, \tilde{\beta}_t I).$$

4. Learned reverse transition

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(\mu_\theta, \sigma_t^2 I).$$

5. Noise parameterization of the reverse mean

$$\mu_\theta = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right).$$

6. Reverse sampling

$$x_{t-1} = \mu_\theta + \sigma_t z.$$

7. Practical training loss

$$L_{\text{simple}} = \mathbb{E} \left[\|\epsilon - \epsilon_{\theta}(x_t, t)\|^2 \right].$$

Chapter 27

Common Conceptual Confusions Resolved

27.1 “The forward process does not train, so is it useless?”

No. It is not trained, but it is essential. It generates x_t and supplies the exact noise target ϵ .

27.2 “Does the network directly output x_{t-1} ?”

Not in the standard noise-prediction formulation. The chain is

$$(x_t, t) \rightarrow \epsilon_\theta \rightarrow \mu_\theta \rightarrow x_{t-1}.$$

27.3 “Is $q(x_{t-1} \mid x_t)$ the training target?”

The tractable posterior used by the variational decomposition is

$$q(x_{t-1} \mid x_t, x_0).$$

The model ultimately operates without x_0 , which is why it learns a reverse transition conditioned on x_t and t .

27.4 “Is the simplified MSE exactly the ELBO?”

No. The exact variational decomposition produces timestep-dependent weighting. The simplified loss removes that weighting:

$$L_{\text{simple}} = \mathbb{E} \|\epsilon - \epsilon_\theta\|^2.$$

27.5 “Why is the reverse step random?”

Because the reverse transition is a probability distribution:

$$x_{t-1} \sim \mathcal{N}(\mu_\theta, \sigma_t^2 I).$$

The random term is therefore part of sampling from the learned model.

27.6 “Why not add all the noise in one step?”

A large corruption step is difficult to reverse. Many small Gaussian perturbations create a sequence of locally easier denoising problems.

27.7 “Why does the network learn images if it predicts noise?”

Because predicting the noise requires distinguishing the noise from the underlying image structure. Across many noise levels, this teaches the network the statistics of valid data.

Chapter 28

A Compact Mental Model

The entire model can be remembered as follows.

FORWARD
 $x_0 \xrightarrow{\text{fixed Gaussian noise}} x_t$
$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$$

Then:

TRAINING
 $(x_t, t) \xrightarrow{\text{NN}} \epsilon_\theta$
$$\text{loss} = \|\epsilon - \epsilon_\theta\|^2$$

Finally:

GENERATION
 $x_T \sim \mathcal{N}(0, I)$
 $x_T \rightarrow x_{T-1} \rightarrow \cdots \rightarrow x_0$
$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z$$

The deepest conceptual statement is:

The network does not directly generate an image.

Instead:

The network learns how to recognize and remove Gaussian noise at every noise level.

Repeated application of this learned denoising operation transforms a random Gaussian sample into a sample from the learned data distribution.

Chapter 29

Conclusion

DDPM combines a deliberately simple fixed stochastic corruption process with a learned reverse process.

The forward process is

$$q(x_t | x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}x_{t-1}, \beta_t I),$$

and its closed-form marginal is

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon.$$

This makes arbitrary noise levels easy to construct during training. The latent-variable structure then permits a variational objective whose central terms match the analytically tractable forward posterior

$$q(x_{t-1} | x_t, x_0)$$

against the learned reverse model

$$p_\theta(x_{t-1} | x_t).$$

The original noise-prediction parameterization converts this variational formulation into the practical regression problem

$$\epsilon_\theta(x_t, t) \approx \epsilon,$$

with simplified objective

$$L_{\text{simple}} = \mathbb{E} \|\epsilon - \epsilon_\theta(x_t, t)\|^2.$$

At generation time, the clean image is no longer available. The model starts from

$$x_T \sim \mathcal{N}(0, I)$$

and repeatedly samples reverse Gaussian transitions until x_0 is obtained.

The same construction can therefore be understood from several mutually reinforcing viewpoints: a latent-variable model, a variational-inference model, a denoising model, a score-learning model, and a Langevin-like stochastic sampling process.

Appendix A

Notation Reference

Symbol	Meaning
x_0	Clean observed data sample.
x_t	Data after t forward diffusion steps.
x_T	Terminal noisy state, approximately standard Gaussian.
t	Diffusion timestep.
T	Total number of diffusion steps.
β_t	Forward noise variance at timestep t .
α_t	$1 - \beta_t$.
$\bar{\alpha}_t$	$\prod_{s=1}^t \alpha_s$.
ϵ	Gaussian noise sampled during training.
ϵ_θ	Neural-network prediction of the noise.
q	Fixed forward diffusion distribution / variational posterior.
p_θ	Learned reverse generative distribution.
$\tilde{\mu}_t$	Exact posterior mean of $q(x_{t-1} \mid x_t, x_0)$.
$\tilde{\beta}_t$	Exact posterior variance coefficient.
μ_θ	Learned reverse-process mean.
σ_t^2	Reverse-process variance coefficient in the fixed-variance formulation.
z	Fresh standard Gaussian noise used during reverse sampling.

Appendix B

Final Summary

1. DDPM defines a fixed forward process that gradually converts data into Gaussian noise.
2. The forward process is a Markov chain.
3. β_t controls the amount of new noise at each step.
4. $\bar{\alpha}_t$ controls how much original signal remains after t steps.
5. The closed-form equation allows direct construction of any x_t from x_0 .
6. DDPM is a latent-variable model with the entire chain $x_{1:T}$ as latent variables.
7. Variational inference supplies a tractable objective because direct likelihood evaluation requires marginalizing over all latent states.
8. The tractable posterior is $q(x_{t-1} \mid x_t, x_0)$.
9. The learned reverse process is $p_\theta(x_{t-1} \mid x_t)$.
10. The standard parameterization predicts the noise $\epsilon_\theta(x_t, t)$.
11. The predicted noise determines the reverse mean.
12. The practical DDPM loss is the MSE between true and predicted noise.
13. During generation, sampling includes Gaussian randomness because the reverse transition is probabilistic.
14. DDPM can also be interpreted through denoising score matching and Langevin-like dynamics.